

Table 1: Common novice programming bug types from selected literature.

BugID	Category	Description	Sources
B1	O	<code>==</code> versus <code>.equals</code> (faulty string comparisons)	[6, 7, 2, 3, 9]
B2	O	<code>=</code> versus <code>==</code>	[7, 2, 3, 14, 11]
B3	I	Misuse <code>{ }</code> , <code>[]</code> , <code>()</code> , <code>" "</code> and <code>' '</code>	[8, 6, 7, 2, 3, 14, 4, 12]
B4	I	Incorrect semi-colon after an <code>if</code> selection structure, before the <code>if</code> statement, or after the <code>for</code> , or <code>while</code> repetition structure, before the respective <code>for</code> or <code>while</code> loop	[6, 7, 2, 3]
B5	I	Wrong separators in <code>for</code> loops (using commas instead of semi-colons)	[7, 2, 3]
B6	I	An <code>if</code> followed by a bracket instead of by a parenthesis	[7, 2, 3]
B7	I	Using keywords as method names or variable names	[7, 2, 3]
B8	I	Invoking methods with wrong arguments	[7, 2, 3, 3, 11, 1, 12]
B9	I	Incorrect semicolon at the end of a method header	[8, 7, 2, 3, 11]
B10	I	Leaving a space after a period when calling a specific method	[7, 2]
B11	I	using <code>=></code> or <code>=<</code> instead of <code>>=</code> and <code><=</code>	[7, 3]
B12	O	Invoking class method on object	[7]
B13	O	Constructor call to non-existent copy constructor	[11, 9]
B14	I	Invoking a non-void method in a statement that requires a return value	[7, 2, 3]
B15	I	Forgetting parentheses after method call	[7, 2, 3, 11]
B16	I	Flow reaches end of non-void method	[7, 2, 3]
B17	O	Misconceptions about object references	[14]
B18	I	Incompatibility between the declared return type of a method and in its invocation	[7, 2, 1, 3, 4, 11]
B19	O	Class declared abstract because of missing function	[7]
B20	O	Trying to invoke a non-static method as if it was static	[2, 3, 11]
B21	O	Field not found	[1]
B22	O	Use of non-static variable inside the static method	[1, 11]
B23	I	Type mismatch	[1, 4, 11, 12]
B24	I	Using a non-initialised variable	[6, 1, 11, 9]
B25	I	Operator missing	[4, 11]
B26	O	Method name not found	[1, 8, 6, 12]
B27	I	Variable not declared	[8, 12, 11, 4, 9]
B28	I	<code>;</code> missing	[8, 12, 4, 11]
B29	I	Variable name written incorrectly	[12, 11]
B30	I	Missing <code>break</code> in <code>switch</code>	[6]
B31	O	Misconceptions related to object arrays	[14]
B32	O	Missing parentheses for constructor call	[12, 11]
B33	O	Unhandled exception	[12, 11]
B34	I	<code>:</code> in place of <code>;</code>	[11]
B35	O	Miconceptions related to inheritance	[14, 9]

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BugID	Category	Description	Sources
B36	O	<code>class</code> or <code>interface</code> expected	[7, 9]
B37	I	Keyword written incorrectly	[11]
B38	O	Misconceptions about access modifiers	[14]
B39	I	Identifier expected	[7]
B40	I	Misspelling <code>println</code>	[7, 8]
B41	O	Misconception related to polymorphic collection	[14]
B42	I	Missing closing curly brace at end of class	[11]
B43	O	Method call targeting wrong type	[11, 9]
B44	I	Not a statement	[7]
B45	O	Class from other package used without <code>import</code>	[11]
B46	I	Pre/Post Increment-Decrement Misuse	[10]
B47	I	Incorrect use of <code>&&</code> vs. <code>&</code> , and <code> </code> vs. <code> </code>	[3]
B48	O	Misunderstanding type casting between objects	[14]
B49	I	Array used in place of array element	[11]
B50	I	Missing closing curly brace after control statement body	[8, 11]
B51	O	Linked list based Queue and Stack implementation errors	[9]
B52	O	Constructor parameter number mismatch	[11, 9]
B53	O	Class not defined	[11, 12]
B54	I	Infix Operator Precedence	[10]
B55	O	Method declaration: no type specified for parameter	[11, 9]
B56	O	Method duplicate exactly	[11]
B57	I	Statement outside method	[11, 9]
B58	I	Comment incorrectly written	[11]
B59	I	Local variable declaration with illegal modifier	[11]
B60	I	Assignment to a collection element	[11]
B61	O	Call to <code>super</code> in constructor not first statement	[11]
B62	O	Collection used in place of element	[11]
B63	O	Constructor call attempted using variable name	[11, 9]
B64	O	Constructor is declared to return <code>void</code>	[11, 6, 9]
B65	I	Method call: missing comma between parameters	[11]
B66	I	Method declaration: semicolon in place of comma	[11]
B67	I	Missing space after <code>new</code>	[11]
B68	O	Constructor call without using <code>new</code>	[11, 9]
B69	O	Custom doubly linked list implementation errors	[9]
B70	O	Value from raw collection must be cast	[11]
B71	I	Variable declaration: name and type in wrong order	[11]
B72	O	Type mismatch: arithmetic performed on String	[11]
B73	I	Misunderstanding of loops	[14]
B74	I	Failing to choose an appropriate loop construct in a specific context	[14]
B75	O	Method declaration: missing method body	[11]
B76	I	<code>for</code> loop is not inclusive so sum excludes the last element	[5]

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BugID	Category	Description	Sources
B77	I	Not realising that division of two integers returns the floor of the division, not a double	[5, 6]
B78	I	Accessing an element outside an array. Usually a result of using <code><=</code> instead of <code><</code>	[5]
B79	O	Class should be declared to implement interface	[11]
B80	O	Using <code>.length</code> as a field	[4]
B81	O	Missing <code>this</code>	[6]
B82	I	Misconceptions in recursion	[13]
B83	O	Mis-capitalized <code>toString</code>	[6]
B84	O	Array-based stack implementation errors	[9]
B85	O	Custom linked list implementation errors	[9]

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